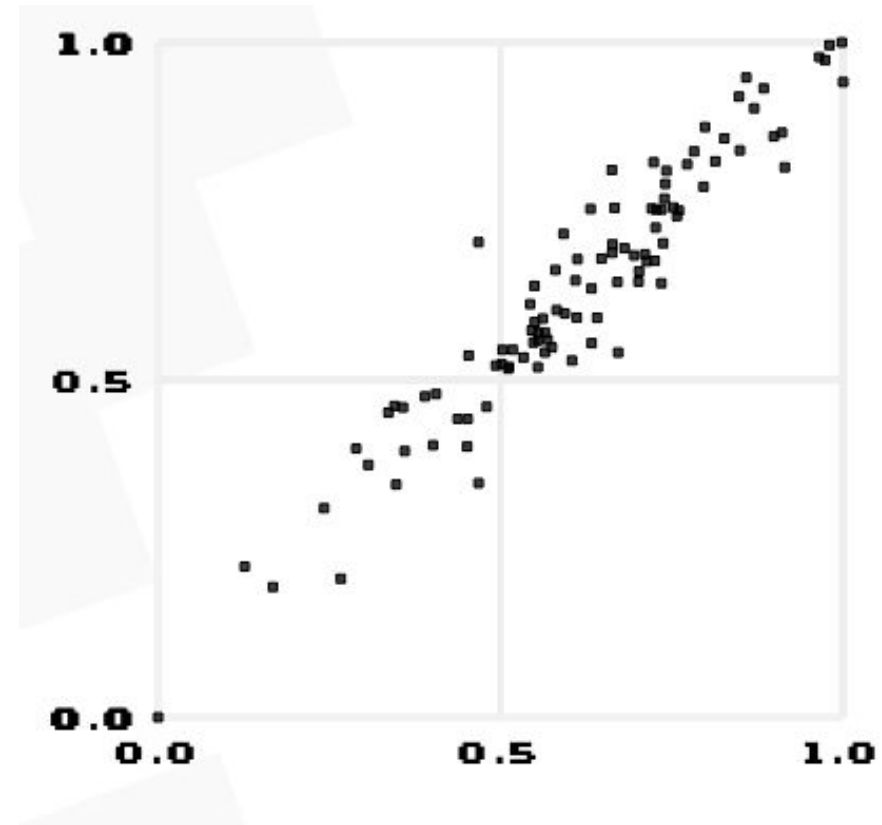
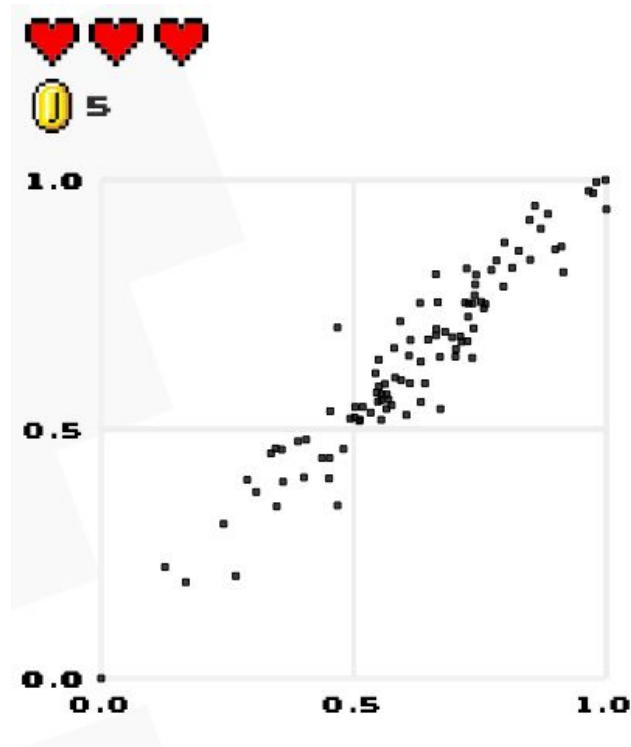




Bayesian Correlation



Correlation



HIGH SCORE 0 **MAIN MENU**

NEXT

TRUE R	0.95
GUESSED R	0.93
DIFFERENCE	0.02
STREAKS	1
MEAN ERROR	0.02

♥ +1 **🪙 +5**

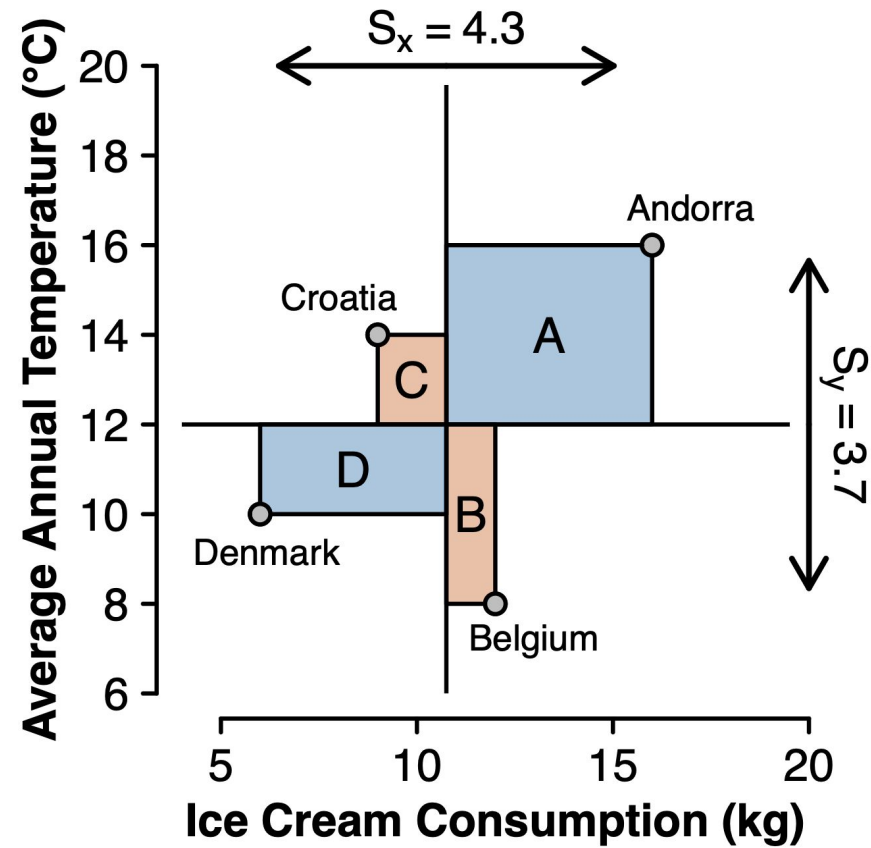
<https://guessthecorrelation.com/>

Correlation

$$r_{xy} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}},$$

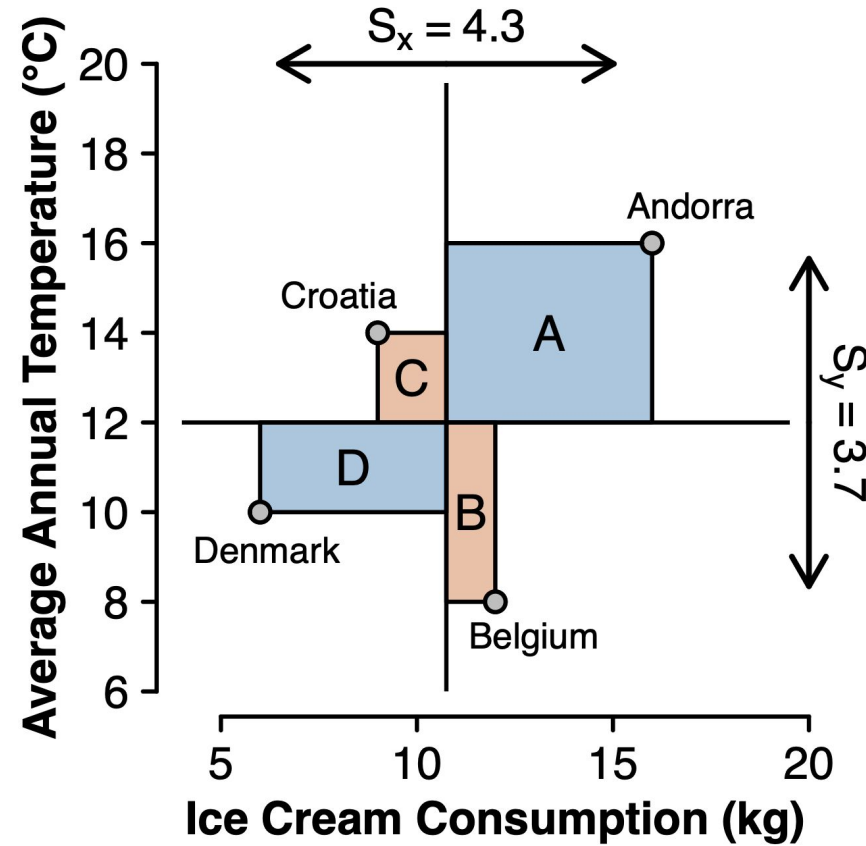
Correlation

$$r_{xy} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}},$$



Correlation

$$r_{xy} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}},$$



... is the difference between concordant and discordant areas, normalized by the total variation

Correlation

- Null hypothesis: The correlation is 0.
“There is no association between two continuous variables”

Correlation

- Null hypothesis: The correlation is 0.
“There is no association between two continuous variables”
- Assumptions: Bivariate normality
 - non-parametric alternatives, for instance for ranks

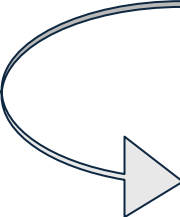
Classical Correlation hypothesis test

$$H_0: \rho = 0 ; H_1: \rho \neq 0$$

Classical Correlation hypothesis test

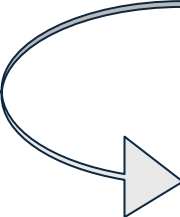
$$H_0: \rho = 0 ; H_1: \rho \neq 0$$

Classical Correlation hypothesis test


$$H_0: \rho = 0 ; H_1: \rho \neq 0$$

$$r_{xy} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}},$$

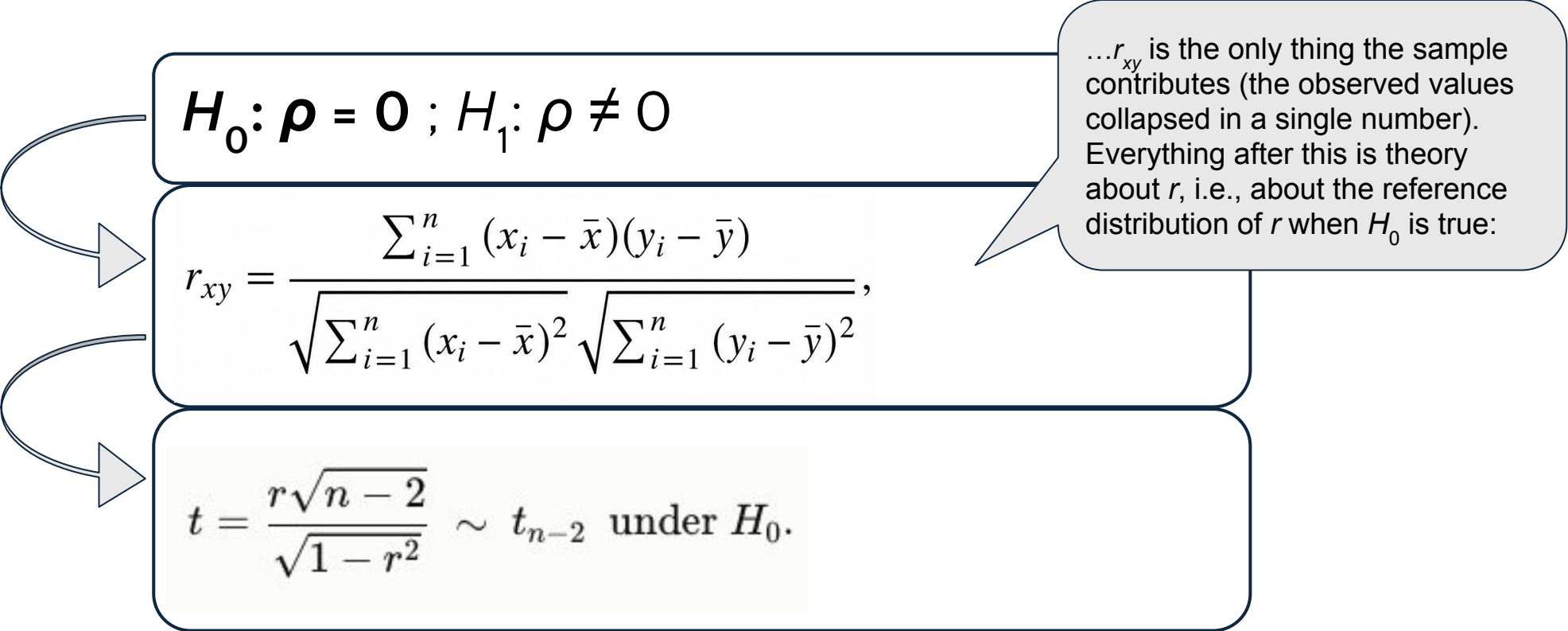
Classical Correlation hypothesis test


$$H_0: \rho = 0 ; H_1: \rho \neq 0$$

$$r_{xy} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}},$$

... r_{xy} is the only thing the sample contributes (the observed values collapsed in a single number). Everything after this is theory about r , i.e., about the reference distribution of r when H_0 is true.

Classical Correlation hypothesis test

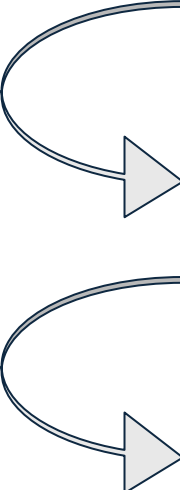

$$H_0: \rho = 0 ; H_1: \rho \neq 0$$

$$r_{xy} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}},$$

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} \sim t_{n-2} \text{ under } H_0.$$

... r_{xy} is the only thing the sample contributes (the observed values collapsed in a single number). Everything after this is theory about r , i.e., about the reference distribution of r when H_0 is true:

Classical Correlation hypothesis test


$$H_0: \rho = 0 ; H_1: \rho \neq 0$$

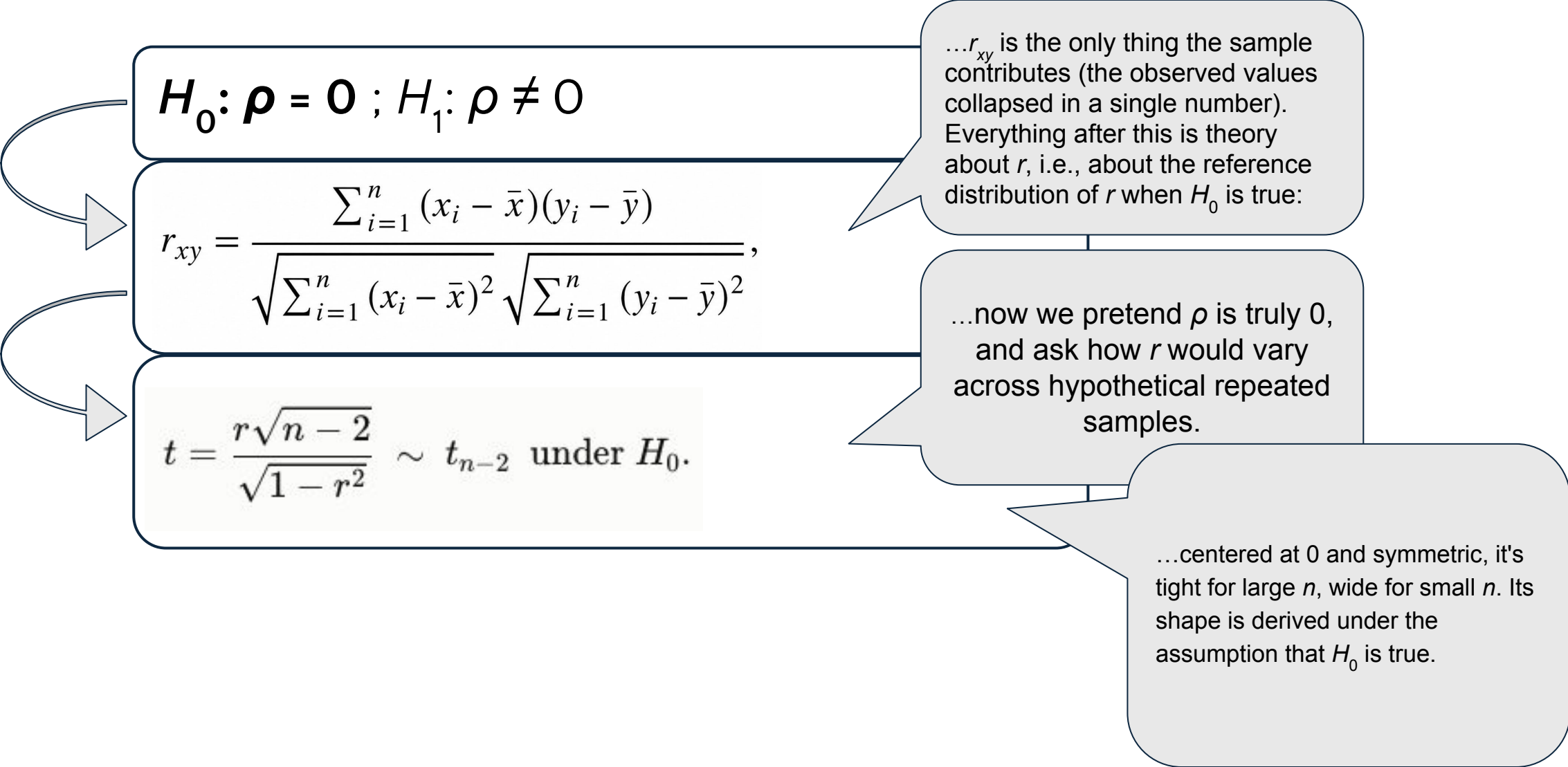
$$r_{xy} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}},$$

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} \sim t_{n-2} \text{ under } H_0.$$

... r_{xy} is the only thing the sample contributes (the observed values collapsed in a single number). Everything after this is theory about r , i.e., about the reference distribution of r when H_0 is true:

...now we pretend ρ is truly 0, and ask how r would vary across hypothetical repeated samples.

Classical Correlation hypothesis test


$$H_0: \rho = 0 ; H_1: \rho \neq 0$$

$$r_{xy} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}},$$

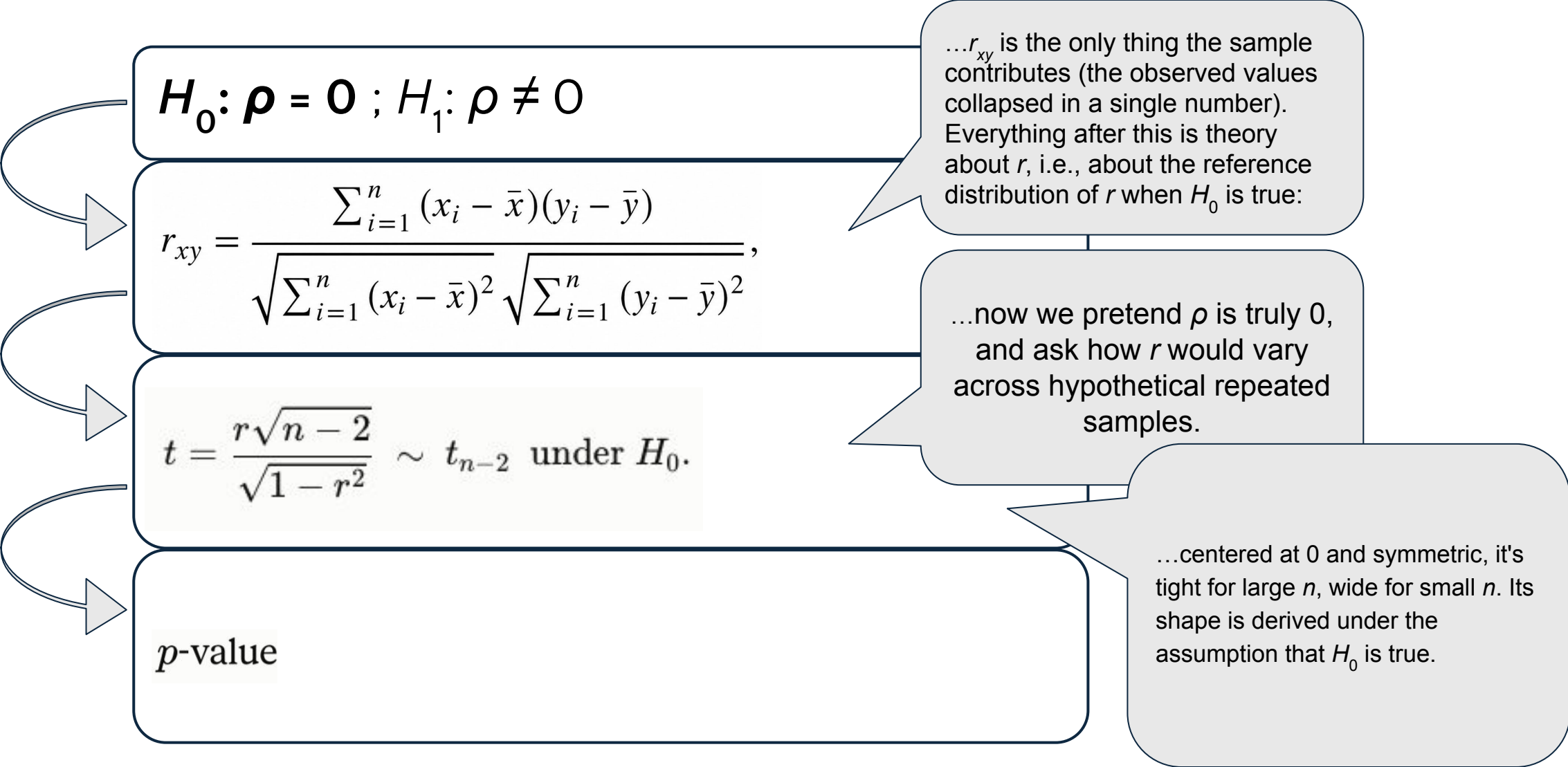
$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} \sim t_{n-2} \text{ under } H_0.$$

... r_{xy} is the only thing the sample contributes (the observed values collapsed in a single number). Everything after this is theory about r , i.e., about the reference distribution of r when H_0 is true:

...now we pretend ρ is truly 0, and ask how r would vary across hypothetical repeated samples.

...centered at 0 and symmetric, it's tight for large n , wide for small n . Its shape is derived under the assumption that H_0 is true.

Classical Correlation hypothesis test


$$H_0: \rho = 0 ; H_1: \rho \neq 0$$

$$r_{xy} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}},$$

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} \sim t_{n-2} \text{ under } H_0.$$

p-value

... r_{xy} is the only thing the sample contributes (the observed values collapsed in a single number). Everything after this is theory about r , i.e., about the reference distribution of r when H_0 is true:

...now we pretend ρ is truly 0, and ask how r would vary across hypothetical repeated samples.

...centered at 0 and symmetric, it's tight for large n , wide for small n . Its shape is derived under the assumption that H_0 is true.

Classical Correlation hypothesis test

$$H_0: \rho = 0 ; H_1: \rho \neq 0$$

$$r_{xy} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}},$$

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} \sim t_{n-2} \text{ under } H_0.$$

p-value

... r_{xy} is the only thing the sample contributes (the observed values collapsed in a single number). Everything after this is theory about r , i.e., about the reference distribution of r when H_0 is true:

...now we pretend ρ is truly 0, and ask how r would vary across hypothetical repeated samples.

...if ρ really were 0, probability of seeing an r at least this extreme just from sampling noise. Compare p to α , **reject or fail to reject**.

...centered at 0 and symmetric, it's tight for large n , wide for small n . Its shape is derived under the assumption that H_0 is true.

Classical Correlation hypothesis test

$$H_0: \rho = 0$$

...frequentist: ρ is a fixed unknown constant; the data (r) are random. The probability statement is about data-given-a-fixed-null.

Classical Correlation hypothesis test

$$H_0: \rho = 0$$

...frequentist: ρ is a fixed unknown constant; the data (r) are random. The probability statement is about data-given-a-fixed-null.

...the whole procedure is built around the single value $\rho = 0$ as the data-generating assumption. **There's no representation at all of "what we'd expect if there were a correlation".**

Classical Correlation hypothesis test

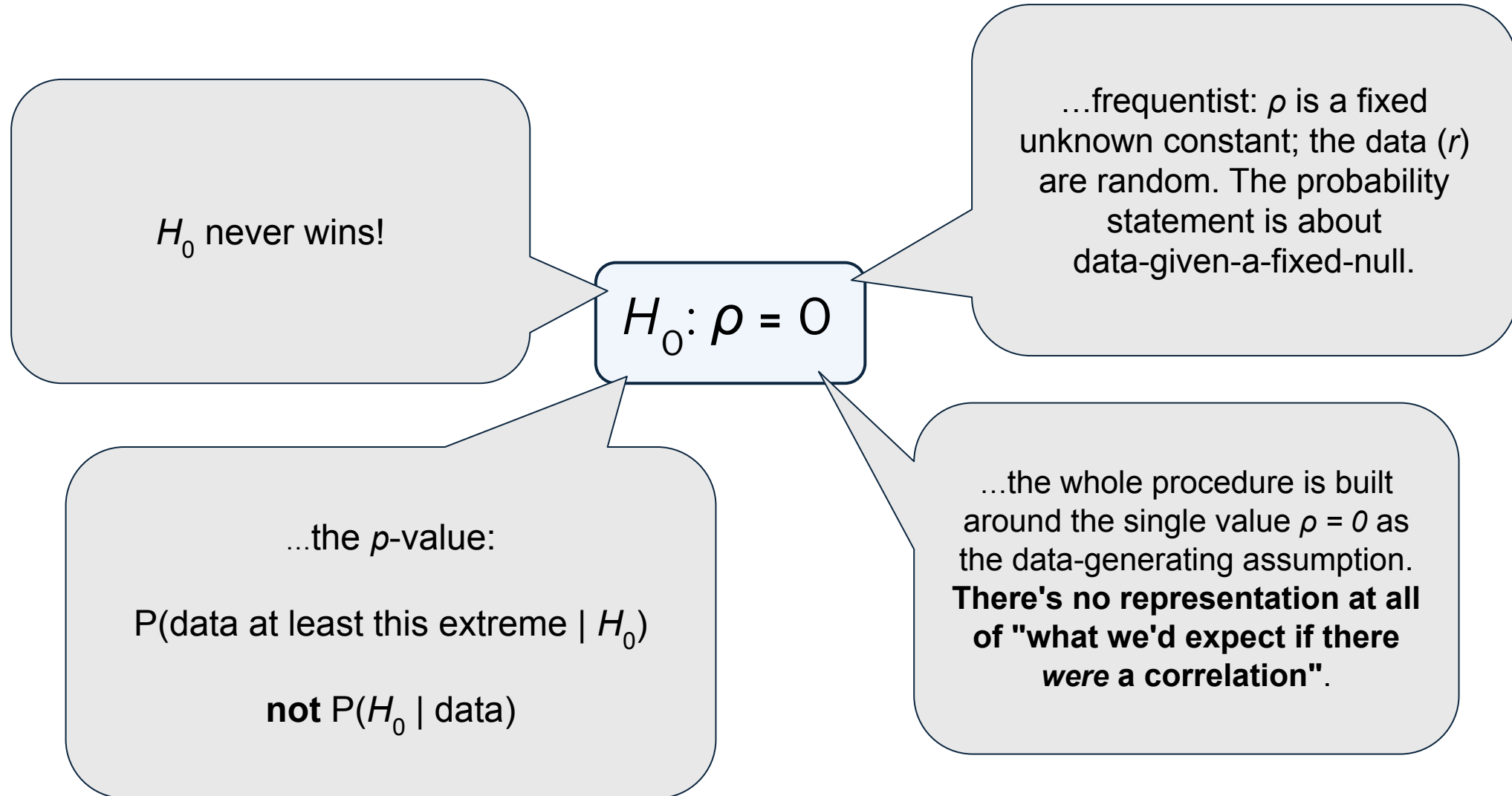
$$H_0: \rho = 0$$

...frequentist: ρ is a fixed unknown constant; the data (r) are random. The probability statement is about data-given-a-fixed-null.

...the p -value:
 $P(\text{data at least this extreme} \mid H_0)$
not $P(H_0 \mid \text{data})$

...the whole procedure is built around the single value $\rho = 0$ as the data-generating assumption.
There's no representation at all of "what we'd expect if there were a correlation".

Classical Correlation hypothesis test



Bayesian Correlation – Theory

$$H_0: \rho = 0$$



$$H_1: \rho \neq 0$$

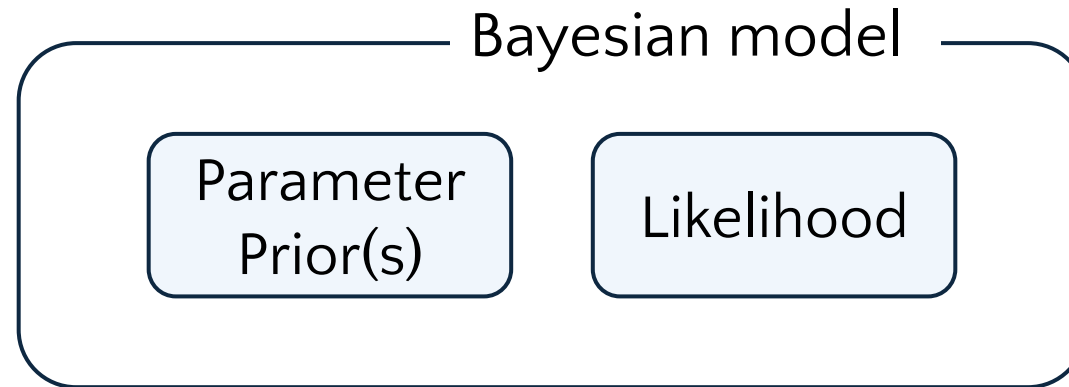
Bayesian Correlation – Theory

$$H_0: \rho = 0 \quad \text{X} \quad H_1: \rho \neq 0$$

$$\underbrace{\frac{P(H_1 | d)}{P(H_0 | d)}}_{\text{posterior odds}} = \underbrace{\frac{P(d | H_1)}{P(d | H_0)}}_{\text{Bayes factor}} \times \underbrace{\frac{P(H_1)}{P(H_0)}}_{\text{prior odds}}.$$

= evidence

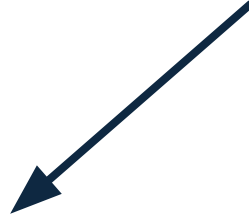
Bayesian Correlation – Theory



Bayesian Correlation – Theory

Likelihood

Density: $f(\text{data} \mid \text{parameters})$



...fix the parameters (say $\mu = 0$, $\sigma = 1$)
and let the data vary: now it tells you
how probable different data outcomes
are. That's a **probability distribution**.

Bayesian Correlation – Theory

Likelihood

Density: $f(\text{data} \mid \text{parameters})$

...fix the parameters (say $\mu = 0$, $\sigma = 1$) and let the data vary: now it tells you how probable different data outcomes are. That's a **probability distribution**.

...fix the data (the study is done, those numbers are locked) and try out different parameter values. The **likelihood** scores each one: how well does this set of parameters explain the data actually observed? Higher = better fit.

Bayesian Correlation – Theory

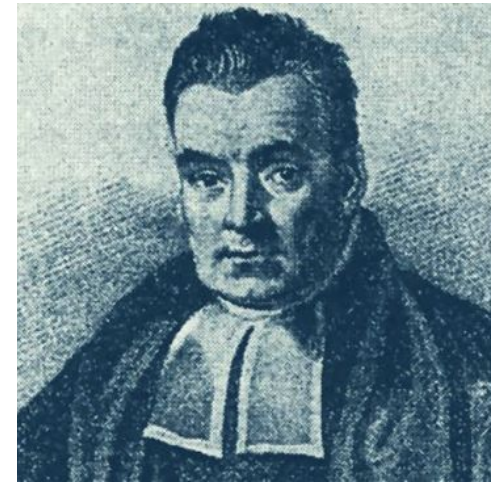
Likelihood

Density: $f(\text{data} \mid \text{parameters})$

...fix the parameters (say $\mu = 0$, $\sigma = 1$) and let the data vary: now it tells you how probable different data outcomes are. That's a **probability distribution**.

...fix the data (the study is done, those numbers are locked) and try out different parameter values. The **likelihood** scores each one: how well does this set of parameters explain the data actually observed? Higher = better fit.

...how good is $\rho = 0.3$?
how good is $\rho = 0.7$?
how good is $\rho = 0$?



Bayesian Correlation – Theory

Likelihood

The *likelihood function* is just the collection of all those output numbers as you vary the guess



Thanks Astra

Bayesian Correlation – Theory

Likelihood

$$\underbrace{L(\theta_0 \mid d)}_{\text{one guess} \rightarrow \text{one score}}$$



Thanks Astra

Bayesian Correlation – Theory

Likelihood

$$\underbrace{L(\theta_0 \mid d)}$$

one guess → one score

$$\underbrace{L(\theta \mid d) \text{ for all } \theta}$$

every guess → a surface of scores



Thanks Astra

Bayesian Correlation – Theory

Likelihood

$$\underbrace{L(\theta_0 \mid d)}$$

one guess → one score

$$\underbrace{L(\theta \mid d) \text{ for all } \theta}$$

every guess → a surface of scores

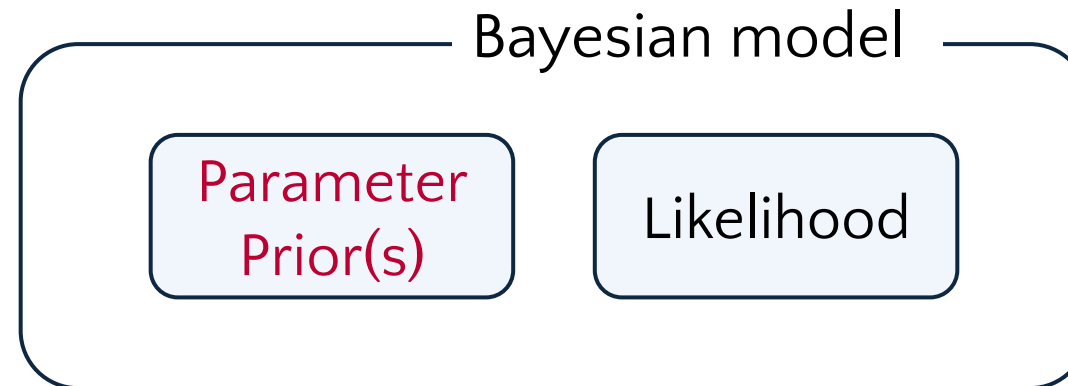
$$\underbrace{P(d \mid H) = \int L(\theta \mid d) p(\theta \mid H) d\theta}$$

prior-weighted average → one number for the model



Thanks Astra

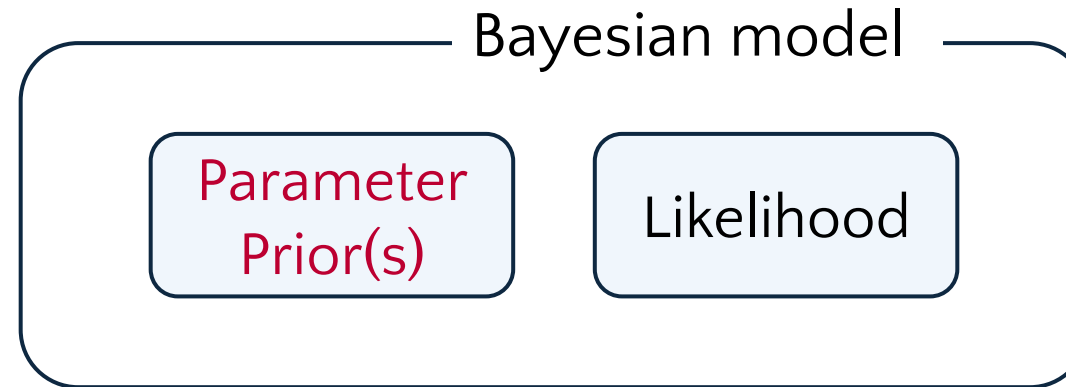
Bayesian Correlation – Theory



$$P(d \mid H) = \int L(\theta \mid d) \boxed{p(\theta \mid H)} d\theta$$

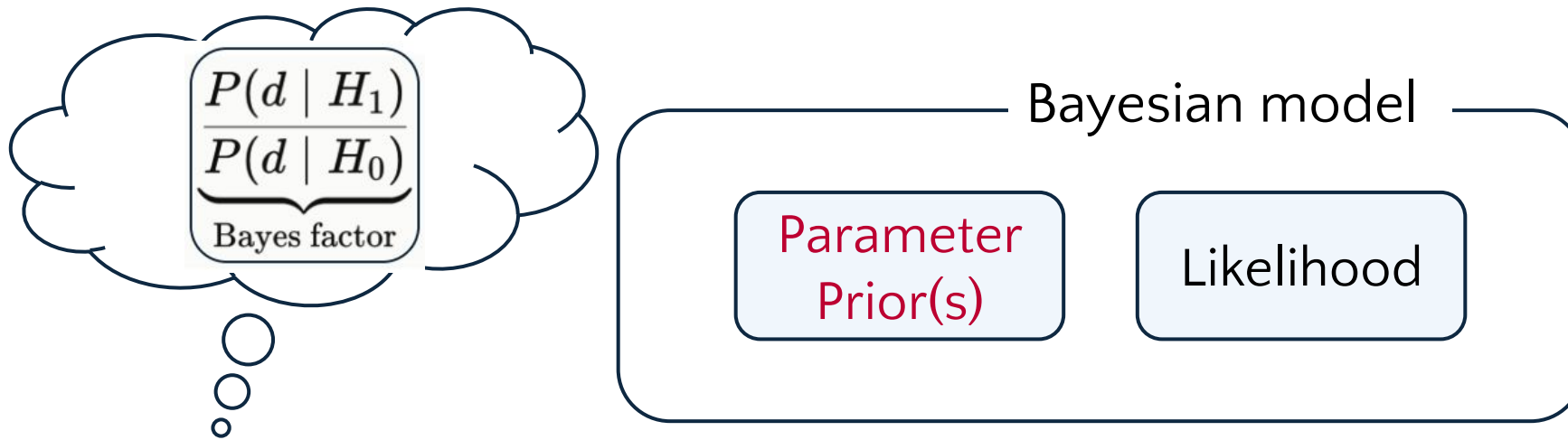
$\underbrace{\hspace{15em}}$
prior-weighted average $\rightarrow$ one number for the model

Bayesian Correlation – Theory



$$\underbrace{P(d \mid H) = \int L(\theta \mid d) p(\theta \mid H) d\theta}_{\text{prior-weighted average} \rightarrow \text{one number for the model}}$$

Bayesian Correlation – Theory



$$P(d | H) = \int L(\theta | d) p(\theta | H) d\theta$$

prior-weighted average → one number for the model

Bayesian Correlation – Theory

Prior(s)

Bayesian Correlation – Theory

Prior(s)

$$H_0: \rho = 0$$

$$\mu_x, \sigma_x, \mu_y, \sigma_y, \rho$$

$$H_1: \rho \neq 0$$

$$\mu_x, \sigma_x, \mu_y, \sigma_y, \rho$$

Bayesian Correlation – Theory

Prior(s)

$$H_0: \rho = 0$$

$$\mu_x, \sigma_x, \mu_y, \sigma_y, \rho$$

$$H_1: \rho \neq 0$$

$$\mu_x, \sigma_x, \mu_y, \sigma_y, \rho$$

Bayesian Correlation – Theory

Prior(s)

$$H_0: \rho = 0$$

$$\mu_x, \sigma_x, \mu_y, \sigma_y$$

$$H_1: \rho \neq 0$$

$$\mu_x, \sigma_x, \mu_y, \sigma_y, \rho$$

Bayesian Correlation – Theory

Prior(s)

$$H_0: \rho = 0$$

$$H_1: \rho \neq 0$$

$$\mu_x, \sigma_x, \mu_y, \sigma_y$$

$$\mu_x, \sigma_x, \mu_y, \sigma_y, \rho$$

$$P(d | H_0) = \int \int \int \int L(\dots, \rho=0) \cdot 1 \cdot 1 \cdot \frac{1}{\sigma_x} \frac{1}{\sigma_y} d\mu_x d\mu_y d\sigma_x d\sigma_y$$

$$P(d | H_1) = \int \int \int \int \int L(\dots, \rho) \cdot 1 \cdot 1 \cdot \frac{1}{\sigma_x} \frac{1}{\sigma_y} \frac{1}{2} d\mu_x d\mu_y d\sigma_x d\sigma_y d\rho$$

Bayesian Correlation – Theory

Prior(s)

$$H_0: \rho = 0$$

$$H_1: \rho \neq 0$$

$$\mu_x, \sigma_x, \mu_y, \sigma_y$$

$$\mu_x, \sigma_x, \mu_y, \sigma_y, \rho$$

$$P(d | H_0) = \int \int \int \int L(\dots, \rho=0) \cdot 1 \cdot 1 \cdot \frac{1}{\sigma_x} \frac{1}{\sigma_y} d\mu_x d\mu_y d\sigma_x d\sigma_y$$

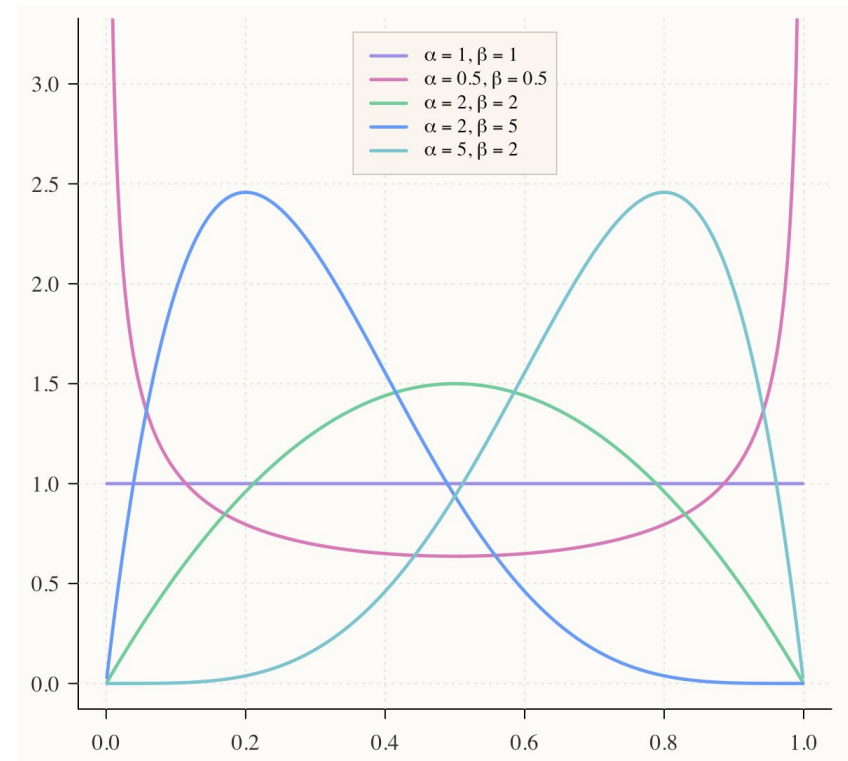
$$P(d | H_1) = \int \int \int \int \int L(\dots, \rho) \cdot 1 \cdot 1 \cdot \frac{1}{\sigma_x} \frac{1}{\sigma_y}^{\frac{1}{2}} d\mu_x d\mu_y d\sigma_x d\sigma_y d\rho$$

Bayesian Correlation – Theory

Prior(s)

$$H_0: \rho = 0$$

$$H_1: \rho \neq 0$$



Bayesian Correlation – Theory

Prior(s)

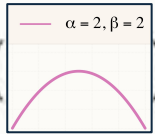
$$H_0: \rho = 0$$

$$\mu_x, \sigma_x, \mu_y, \sigma_y$$

$$H_1: \rho \neq 0$$

$$\mu_x, \sigma_x, \mu_y, \sigma_y, \rho$$

$$P(d | H_0) = \int \int \int \int L(\dots, \rho=0) \cdot 1 \cdot 1 \cdot \frac{1}{\sigma_x} \frac{1}{\sigma_y} d\mu_x d\mu_y d\sigma_x d\sigma_y$$

$$P(d | H_1) = \int \int \int \int \int L(\text{) \cdot 1 \cdot 1 \cdot \frac{1}{\sigma_x} \frac{1}{\sigma_y} \frac{1}{2} d\mu_x d\mu_y d\sigma_x d\sigma_y d\rho$$

Bayesian Correlation – Theory

$$H_0: \rho = 0$$

$$\mu_x, \sigma_x, \mu_y, \sigma_y$$



$$H_1: \rho \neq 0$$

$$\mu_x, \sigma_x, \mu_y, \sigma_y, \rho$$

$$P(d \mid H_1) = \int \int \int \int \int L(\text{---} \alpha=2, \beta=2 \text{---}) \cdot 1 \cdot 1 \cdot \frac{1}{\sigma_x} \frac{1}{\sigma_y} \frac{1}{2} d\mu_x d\mu_y d\sigma_x d\sigma_y d\rho$$

$$P(d \mid H_0) = \int \int \int \int L(\dots, \rho=0) \cdot 1 \cdot 1 \cdot \frac{1}{\sigma_x} \frac{1}{\sigma_y} d\mu_x d\mu_y d\sigma_x d\sigma_y$$

Bayesian Correlation – Theory

$$H_0: \rho = 0$$

$$\mu_x, \sigma_x, \mu_y, \sigma_y$$



$$H_1: \rho \neq 0$$

$$\mu_x, \sigma_x, \mu_y, \sigma_y, \rho$$

$$P(d | H_1) = \int \int \int \int \int L(\text{---} \alpha = 2, \beta = 2) \cdot 1 \cdot 1 \cdot \frac{1}{\sigma_x} \frac{1}{\sigma_y} \frac{1}{2} d\mu_x d\mu_y d\sigma_x d\sigma_y d\rho$$

$$P(d | H_0) = \int \int \int \int L(\dots, \rho=0) \cdot 1 \cdot 1 \cdot \frac{1}{\sigma_x} \frac{1}{\sigma_y} d\mu_x d\mu_y d\sigma_x d\sigma_y$$

$$\frac{P(d | H_1)}{P(d | H_0)} = \underbrace{\text{Bayes factor}}_{= \text{evidence}} = \text{BF}_{10}$$

Bayesian Correlation – Theory

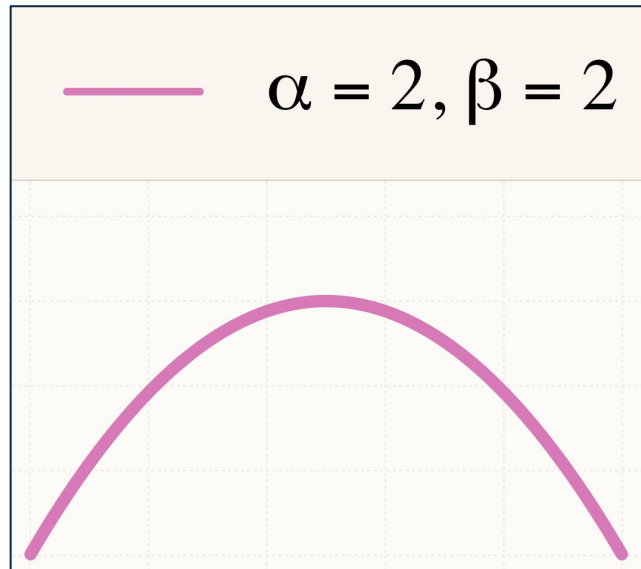
$$\underbrace{\frac{P(d | H_1)}{P(d | H_0)}}_{\text{Bayes factor}} = \text{BF}_{10}$$

= evidence

Testing (Bayes factor): "Is there an effect at all?":

- compares $H_0 : \rho = 0$ against $H_1 : \rho \neq 0$.
- Output: a number weighing one model against the other.

Bayesian Correlation – Theory

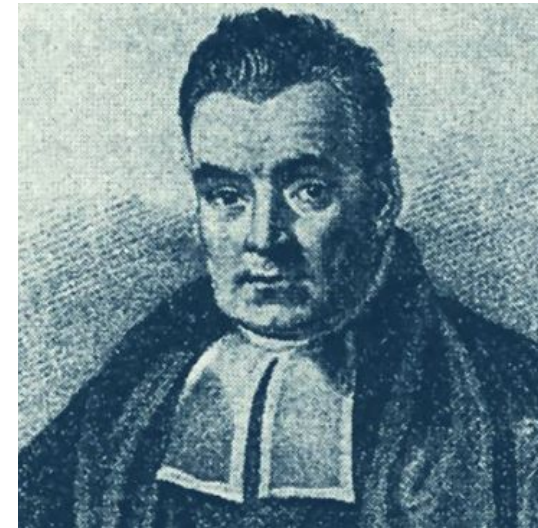


Estimation (posterior): "Assuming there is an effect, how big is it, and how sure are we?"

- entirely based on H_1
- Output: a whole updated distribution over ρ

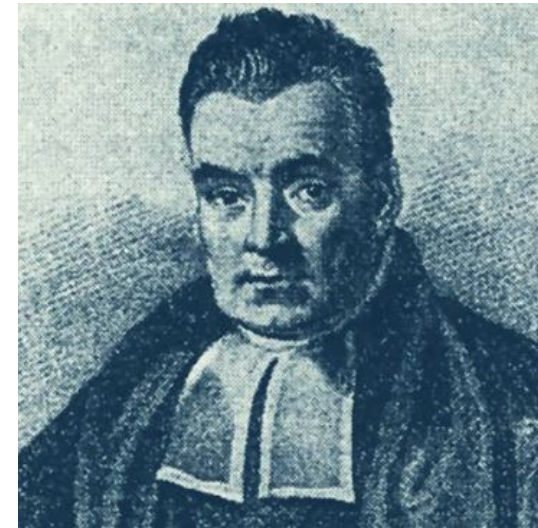
Bayesian Correlation – Theory

$$\underbrace{p(\theta \mid d)}_{\text{posterior}} = \frac{\overbrace{L(\theta \mid d)}^{\text{likelihood}} \overbrace{p(\theta \mid H_1)}^{\text{prior}}}{\underbrace{\int L(\theta \mid d) p(\theta \mid H_1) d\theta}_{\text{normalizing constant} = P(d|H_1)}}$$



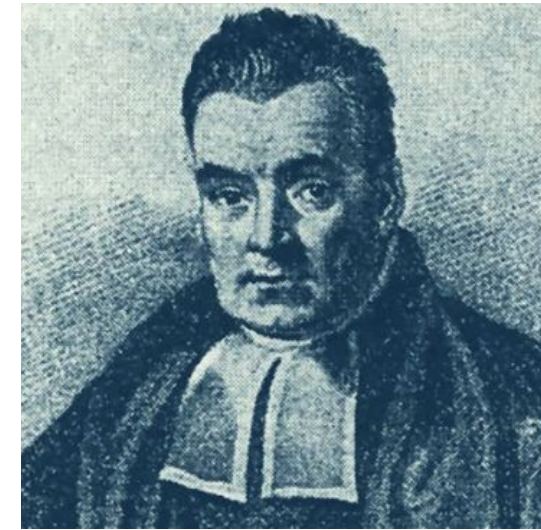
Bayesian Correlation – Theory

$$\underbrace{p(\theta \mid d)}_{\text{posterior}} = \frac{\overbrace{L(\theta \mid d)}^{\text{likelihood}} \overbrace{p(\theta \mid H_1)}^{\text{prior}}}{\underbrace{\int L(\theta \mid d) p(\theta \mid H_1) d\theta}_{\text{normalizing constant} = P(d|H_1)}}$$



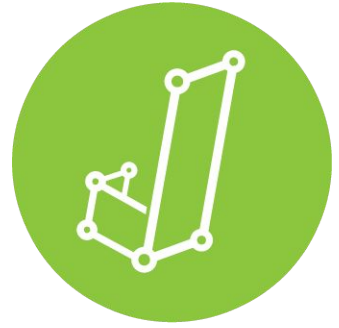
Bayesian Correlation – Theory

$$\underbrace{p(\theta \mid d)}_{\text{posterior}} = \frac{\overbrace{L(\theta \mid d)}^{\text{likelihood}} \overbrace{p(\theta \mid H_1)}^{\text{prior}}}{\underbrace{\int L(\theta \mid d) p(\theta \mid H_1) d\theta}_{\text{normalizing constant} = P(d|H_1)}}$$



$$\underbrace{p(\rho \mid d)}_{\text{posterior for the effect}} = \int \underbrace{p(\theta \mid d)}_{\text{joint posterior}} d\eta \quad \text{where } \eta = (\mu_x, \mu_y, \sigma_x, \sigma_y)$$

Bayesian Correlation – Example Part 1

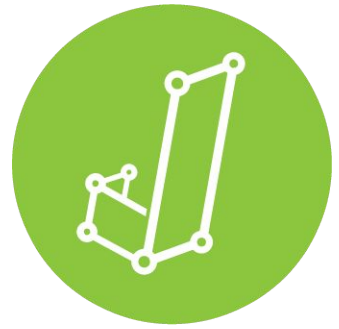


Let's do a **Bayesian Correlation analysis** together.

Example data set:

Adam Sandler (JASP data library)

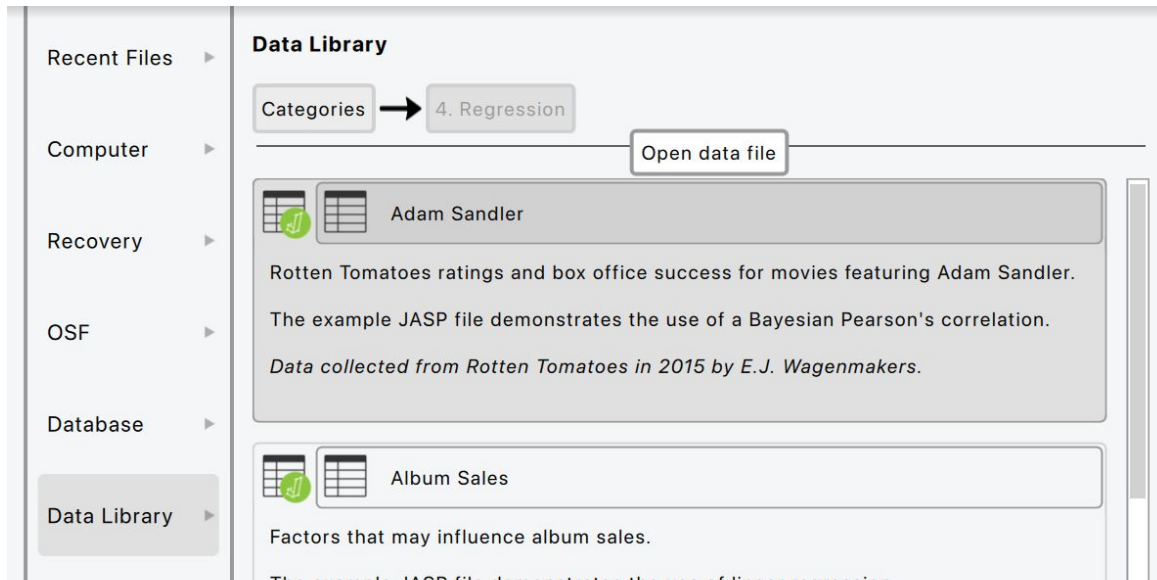
Bayesian Correlation – Example Part 1



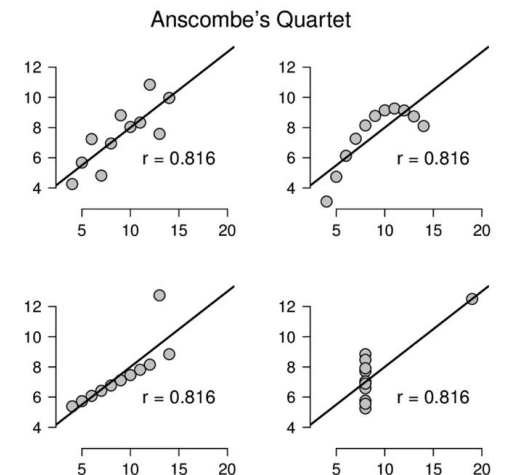
Let's do a **Bayesian Correlation analysis** together.

Example data set:

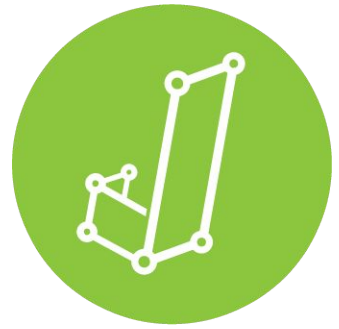
Adam Sandler (JASP data library)



P.S.: Always plot your data!



Bayesian Correlation – Exercise



Your turn!

Exercise data set: **Do lonely people take hotter showers?**

use showers.csv

Feel free to first do a classical analysis.
Then, please do a Bayesian reanalysis!



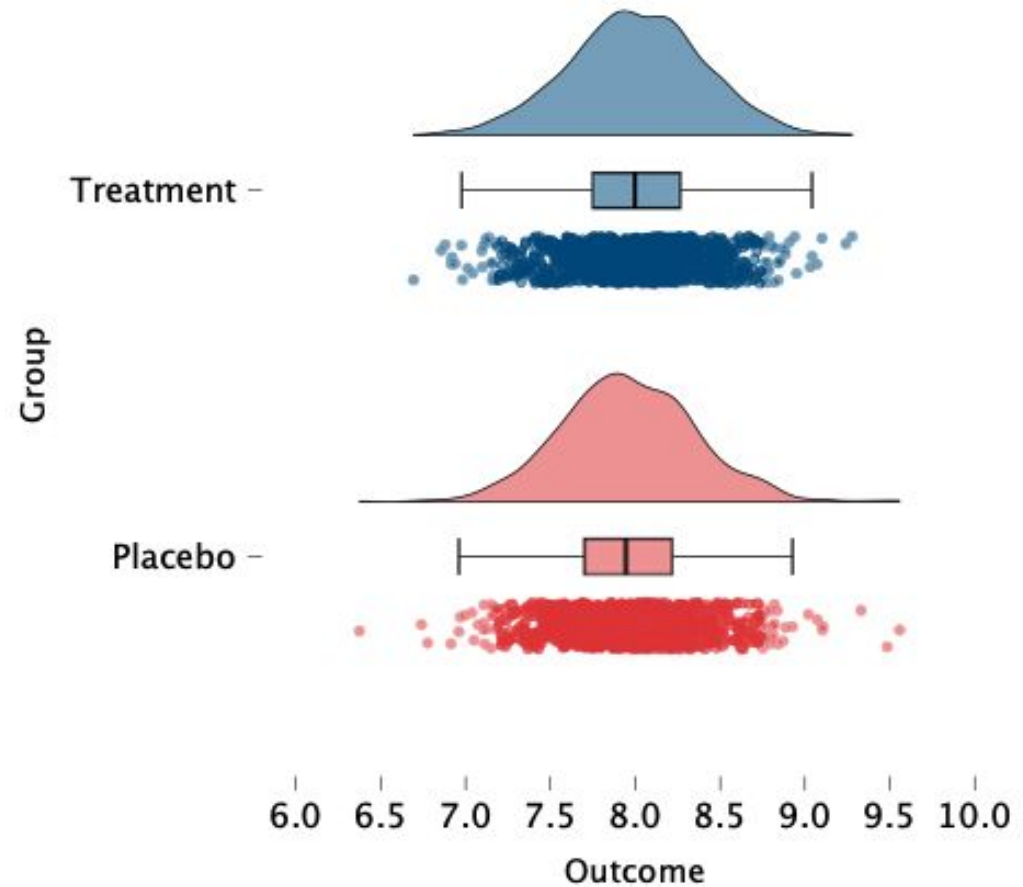
Bayesian Correlation – Further Reading

Wagenmakers, E. J., Godmann, H., De Ruiter, J., & van Doorn, J. (2023). Visualizing the Equation for the Sample Correlation Coefficient.

Wagenmakers, E. J., Verhagen, J., & Ly, A. (2016). How to quantify the evidence for the absence of a correlation. *Behavior Research Methods*, 48(2), 413-426. **!**



Bayesian t -Test



t -Test

- Most commonly: Is there a difference between two groups on a continuous outcome measure?

t -Test

- Most commonly: Is there a difference between two groups on a continuous outcome measure? -> **Independent samples t -test**

t -Test

- Most commonly: Is there a difference between two groups on a continuous outcome measure? -> **Independent samples t -test**
- Others:
 - One-sample t -test
 - Paired t -test

t -Test

- Most commonly: Is there a difference between two groups on a continuous outcome measure? -> **Independent samples t -test**
- Secondary question: If there is an effect, how large is it?

t -Test

- Most commonly: Is there a difference between two groups on a continuous outcome measure? -> **Independent samples t -test**
- Secondary question: If there is an effect, how large is it?
 - Sample effect size -> Cohen's d (difference in means / common sd)

t-Test

- Most commonly: Is there a difference between two groups on a continuous outcome measure? -> **Independent samples *t*-test**
- Secondary question: If there is an effect, how large is it?
 - Sample effect size -> Cohen's *d* (difference in means / common sd)
 - Population version: $\delta = \mu / \sigma$

Classical t -Test – Theory

group 1 $\sim \mathcal{N}(\mu_1, \sigma^2)$, group 2 $\sim \mathcal{N}(\mu_2, \sigma^2)$.

$\delta = \mu_1 - \mu_2$, $H_0 : \delta = 0$ vs $H_1 : \delta \neq 0$.

- Assumptions:
 - Both groups drawn from independent Normal distributions
 - Shared variance (other exist, i.e., Welch's t -test)

Classical t -Test – Theory

$$\bar{x}_1 - \bar{x}_2$$

$$t = \frac{\text{signal}}{\text{noise}} = \frac{\text{observed gap between groups}}{\text{how much that gap would wobble by chance}}$$

$$s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

Two-sided ($H_1 : \delta \neq 0$): both tails,
 $p = P(|T_\nu| \geq |t_{\text{obs}}|) = 2 P(T_\nu \geq |t_{\text{obs}}|)$

Classical t -Test – Theory

- p is $P(\text{data at least this extreme} \mid H_0)$, **not** $P(H_0 \mid \text{data})$

Classical t -Test – Theory

- p is P (data at least this extreme $| H_0$), **not** P (H_0 $|$ data)
- You can reject H_0 , but you can never accept it: "not significant" is not "the groups are equal"

Classical t -Test – Theory

- p is $P(\text{data at least this extreme} \mid H_0)$, **not** $P(H_0 \mid \text{data})$
- You can reject H_0 , but you can never accept it: "not significant" is not "the groups are equal"
- The test does not quantify evidence: A tiny, trivial gap becomes significant with enough observations

Two-sided ($H_1 : \delta \neq 0$): both tails,

$$p = P(|T_\nu| \geq |t_{\text{obs}}|) = 2 P(T_\nu \geq |t_{\text{obs}}|)$$

Bayesian t -Test – Theory

Model

$$\text{group 1} \sim \mathcal{N}(\mu_1, \sigma^2), \quad \text{group 2} \sim \mathcal{N}(\mu_2, \sigma^2).$$

$$\delta = \mu_1 - \mu_2, \quad H_0 : \delta = 0 \quad \text{vs} \quad H_1 : \delta \neq 0.$$

Bayesian t -Test – Theory

Model

$$\text{group 1} \sim \mathcal{N}(\mu_1, \sigma^2), \quad \text{group 2} \sim \mathcal{N}(\mu_2, \sigma^2).$$

$$\delta = \mu_1 - \mu_2, \quad H_0 : \delta = 0 \quad \text{vs} \quad H_1 : \delta \neq 0.$$

Bayesian t -Test – Theory

Model

$$x_1 \sim \mathcal{N}\left(\mu + \frac{\alpha}{2}, \sigma^2\right)$$

$$x_2 \sim \mathcal{N}\left(\mu - \frac{\alpha}{2}, \sigma^2\right)$$

$$\text{group 1} \sim \mathcal{N}(\mu_1, \sigma^2), \quad \text{group 2} \sim \mathcal{N}(\mu_2, \sigma^2).$$

$$\delta = \mu_1 - \mu_2, \quad H_0 : \delta = 0 \quad \text{vs} \quad H_1 : \delta \neq 0.$$

Bayesian t -Test – Theory

Model

$$x_1 \sim \mathcal{N}\left(\mu + \frac{\alpha}{2}, \sigma^2\right)$$

$$x_2 \sim \mathcal{N}\left(\mu - \frac{\alpha}{2}, \sigma^2\right)$$

$$\text{group 1} \sim \mathcal{N}(\mu_1, \sigma^2), \quad \text{group 2} \sim \mathcal{N}(\mu_2, \sigma^2).$$

$$\delta = \mu_1 - \mu_2, \quad H_0 : \delta = 0 \quad \text{vs} \quad H_1 : \delta \neq 0.$$

Bayesian t -Test – Theory

Model

$$x_1 \sim \mathcal{N}\left(\mu + \frac{\alpha}{2}, \sigma^2\right)$$

$$x_2 \sim \mathcal{N}\left(\mu - \frac{\alpha}{2}, \sigma^2\right)$$

$$\delta = \frac{\alpha}{\sigma}$$

Bayesian t -Test – Theory

Prior(s)

$$x_1 \sim \mathcal{N}\left(\mu + \frac{\alpha}{2}, \sigma^2\right)$$

$$x_2 \sim \mathcal{N}\left(\mu - \frac{\alpha}{2}, \sigma^2\right)$$

$$\delta = \frac{\alpha}{\sigma}$$

Bayesian t -Test – Theory

Prior(s)

$$x_1 \sim \mathcal{N}(\mu + \frac{\alpha}{2}, \sigma^2)$$

$$x_2 \sim \mathcal{N}(\mu - \frac{\alpha}{2}, \sigma^2)$$

$$\delta = \frac{\alpha}{\sigma}$$

Bayesian t -Test – Theory

Prior(s)

$$x_1 \sim \mathcal{N}(\mu + \frac{\alpha}{2}, \sigma^2)$$

$$x_2 \sim \mathcal{N}(\mu - \frac{\alpha}{2}, \sigma^2)$$

$$\delta = \frac{\alpha}{\sigma}$$

Bayesian t -Test – Theory

Prior(s)

$$x_1 \sim \mathcal{N}(\mu + \frac{\alpha}{2}, \sigma^2)$$

$$x_2 \sim \mathcal{N}(\mu - \frac{\alpha}{2}, \sigma^2)$$

$$\delta = \frac{\alpha}{\sigma}$$

The test-irrelevant parameters μ and σ are common to all of the models. These parameters can be assigned prior distributions that are *weakly informed by the observed data...*

(important to remember for hands-on modeling: variance parameters cancel when they only set the scale of the noise; they distort the effect when they also decide how much of the data counts as noise)

Bayesian t -Test – Theory

Prior(s)

$$x_1 \sim \mathcal{N}\left(\mu + \frac{\alpha}{2}, \sigma^2\right)$$

$$x_2 \sim \mathcal{N}\left(\mu - \frac{\alpha}{2}, \sigma^2\right)$$

$$\boxed{\delta} = \frac{\alpha}{\sigma}$$

Bayesian t -Test – Theory

Prior(s)

Bayesian t -Test – Theory

Prior(s)

Subjective

Objective

Bayesian t -Test – Theory

Prior(s)

Subjective

- The literature suggests the kinds of effect sizes that are plausible

Objective

Bayesian t -Test – Theory

Prior(s)

Subjective

- The literature suggests the kinds of effect sizes that are plausible
- Earlier experiments on similar topics may give more specific information

Objective

Bayesian t -Test – Theory

Prior(s)

Subjective

- The literature suggests the kinds of effect sizes that are plausible
- Earlier experiments on similar topics may give more specific information
- Expert knowledge yields relatively precise predictions

Objective

Bayesian t -Test – Theory

Prior(s)

Subjective

- The literature suggests the kinds of effect sizes that are plausible
- Earlier experiments on similar topics may give more specific information
- Expert knowledge yields relatively precise predictions
- Drawback: effortful and “subjective”

Objective

Bayesian t -Test – Theory

Prior(s)

Subjective

- The literature suggests the kinds of effect sizes that are plausible
- Earlier experiments on similar topics may give more specific information
- Expert knowledge yields relatively precise predictions
- Drawback: effortful and “subjective”

Objective

- Pick the prior (or family of priors) that meets a series of desiderata

Bayesian t -Test – Theory

Prior(s)

Subjective

- The literature suggests the kinds of effect sizes that are plausible
- Earlier experiments on similar topics may give more specific information
- Expert knowledge yields relatively precise predictions
- Drawback: effortful and “subjective”

Objective

- Pick the prior (or family of priors) that meets a series of desiderata
- Advantage: can be applied in any scenario, forms a point of reference for a more detailed analysis, “objective”

Bayesian t -Test – Theory

Prior(s)

Objective – desiderata

Bayesian t -Test – Theory

Prior(s)

Objective – desiderata

- Effect size ranges from $-\infty$ to $+\infty$, and so should the prior

Bayesian t -Test – Theory

Prior(s)

Objective – desiderata

- Effect size ranges from $-\infty$ to $+\infty$, and so should the prior
- The fact that a test is conducted presupposes that the effect is fairly small

Bayesian t -Test – Theory

Prior(s)

Objective – desiderata

- Effect size ranges from $-\infty$ to $+\infty$, and so should the prior
- The fact that a test is conducted presupposes that the effect is fairly small
- Consistency (as sample size grows, probability of preferring the correct model goes to 1)

Bayesian t -Test – Theory

Prior(s)

Objective – desiderata

- Effect size ranges from $-\infty$ to $+\infty$, and so should the prior
- The fact that a test is conducted presupposes that the effect is fairly small
- Consistency (as sample size grows, probability of preferring the correct model goes to 1)
- Completely uninformative data should give a Bf of 1

Bayesian t -Test – Theory

Prior(s)

Objective – desiderata

- Effect size ranges from $-\infty$ to $+\infty$, and so should the prior
- The fact that a test is conducted presupposes that the effect is fairly small
- Consistency (as sample size grows, probability of preferring the correct model goes to 1)
- Completely uninformative data should give a Bf of 1
- Completely informative data should yield a Bf of infinity

Bayesian t -Test – Theory

Prior(s)

Objective – desiderata

Candidates

- Effect size ranges from $-\infty$ to $+\infty$, and so should the prior
- The fact that a test is conducted presupposes that the effect is fairly small
- Consistency (as sample size grows, probability of preferring the correct model goes to 1)
- Completely uninformative data should give a Bf of 1
- Completely informative data should yield a Bf of infinity

Bayesian t -Test – Theory

Prior(s)

Objective – desiderata

- Effect size ranges from $-\infty$ to $+\infty$, and so should the prior
- The fact that a test is conducted presupposes that the effect is fairly small
- Consistency (as sample size grows, probability of preferring the correct model goes to 1)
- Completely uninformative data should give a Bf of 1
- Completely informative data should yield a Bf of infinity

Candidates

- Normal?

Bayesian t -Test – Theory

Prior(s)

Objective – desiderata

- Effect size ranges from $-\infty$ to $+\infty$, and so should the prior
- The fact that a test is conducted presupposes that the effect is fairly small
- Consistency (as sample size grows, probability of preferring the correct model goes to 1)
- Completely uninformative data should give a Bf of 1
- Completely informative data should yield a Bf of infinity

Candidates

- Normal?

Bayesian t -Test – Theory

Prior(s)

Objective – desiderata

- Effect size ranges from $-\infty$ to $+\infty$, and so should the prior
- The fact that a test is conducted presupposes that the effect is fairly small
- Consistency (as sample size grows, probability of preferring the correct model goes to 1)
- Completely uninformative data should give a Bf of 1
- Completely informative data should yield a Bf of infinity

Candidates

- Normal?
- Uniform?

Bayesian t -Test – Theory

Prior(s)

Objective – desiderata

- Effect size ranges from $-\infty$ to $+\infty$, and so should the prior
- The fact that a test is conducted presupposes that the effect is fairly small
- Consistency (as sample size grows, probability of preferring the correct model goes to 1)
- Completely uninformative data should give a Bf of 1
- Completely informative data should yield a Bf of infinity

Candidates

- Normal?
- Uniform?

Bayesian t -Test – Theory

Prior(s)

Objective – desiderata

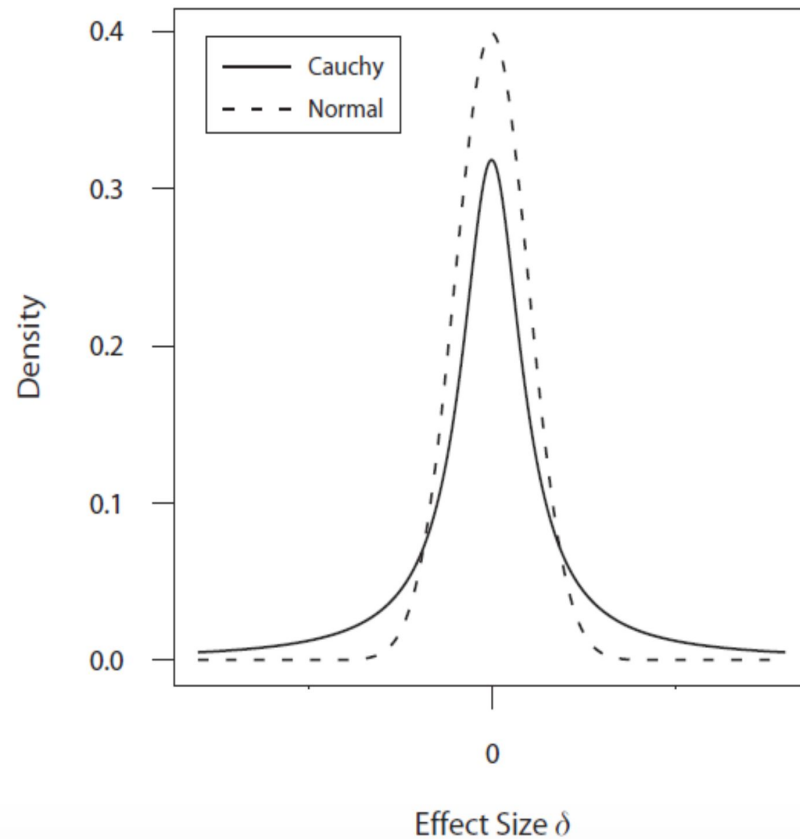
- Effect size ranges from $-\infty$ to $+\infty$, and so should the prior
- The fact that a test is conducted presupposes that the effect is fairly small
- Consistency (as sample size grows, probability of preferring the correct model goes to 1)
- Completely uninformative data should give a Bf of 1
- Completely informative data should yield a Bf of infinity

Candidates

- Normal?
- Uniform?
- Cauchy!

Bayesian t -Test – Theory

Prior(s)



Candidates

- Normal?
- Uniform?
- Cauchy!

Bayesian t -Test – Theory

$$H_0 : \delta = 0 \quad H_1 : \delta \sim \text{Cauchy}\left(0, \frac{1}{\sqrt{2}}\right)$$

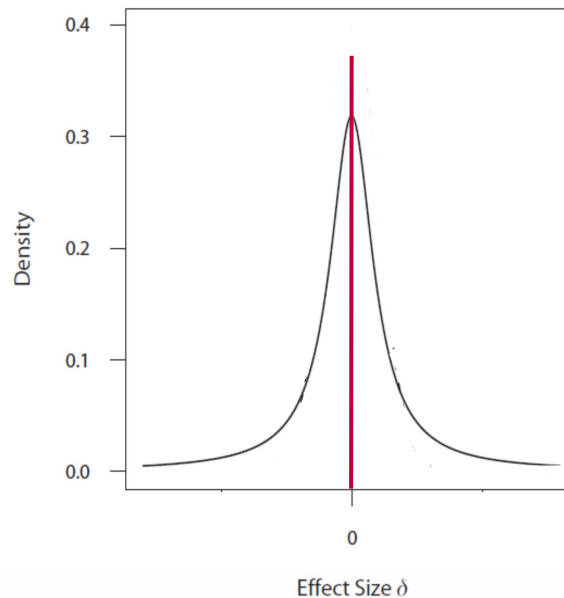
Bayesian t -Test – Theory

$$x_1 \sim \mathcal{N}\left(\mu + \frac{\alpha}{2}, \sigma^2\right)$$

$$x_2 \sim \mathcal{N}\left(\mu - \frac{\alpha}{2}, \sigma^2\right)$$

$$\delta = \frac{\alpha}{\sigma}$$

$$H_0 : \delta = 0 \quad H_1 : \delta \sim \text{Cauchy}\left(0, \frac{1}{\sqrt{2}}\right)$$



Bayesian t -Test – Theory

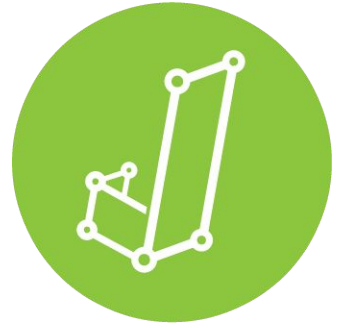
Testing:

$$\underbrace{\frac{P(d \mid H_1)}{P(d \mid H_0)}}_{\text{Bayes factor}} = \text{evidence}$$

Estimation:

$$\underbrace{p(\delta \mid d)}_{\text{posterior for the effect}} = \int \underbrace{p(\theta \mid d)}_{\text{joint posterior}} d\eta \quad \text{where } \eta = (\mu, \sigma)$$

Bayesian t -Test – Example



Let's do a **Bayesian t -test analysis** together.

Example data set:

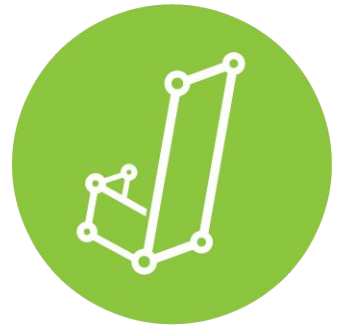
Turning the Wheels of Time

→ JASP data library
(Kitchen Rolls under t -test)



Topolinski, S., & Sparenberg, P. (2012). Turning the hands of time: Clockwise movements increase preference for novelty. *Social Psychological and Personality Science*, 3(3), 308-314.

Bayesian t -Test – Exercise



Your turn!

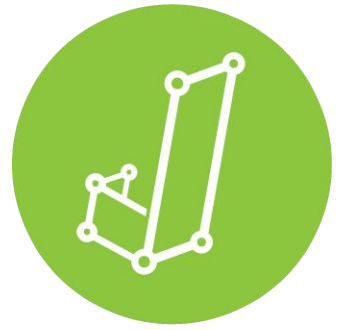
Exercise data set:

use `insomnia_data.csv`

Feel free to first do a classical analysis.
Then, please do a Bayesian reanalysis :)



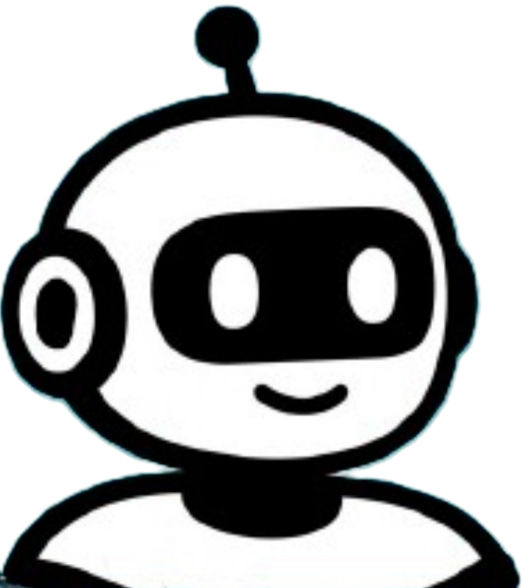
Bayesian t -Test – Exercise



Your turn!

Exercise data set:

use `insomnia_data.csv`



Bayesian t -Test – Further Reading

Wagenmakers, E. J., Verhagen, J., & Ly, A. (2016). How to quantify the evidence for the absence of a correlation. *Behavior Research Methods*, 48(2), 413-426.